

Roll No.

TPE-002

B. TECH. (ECE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
OPTICAL FIBER COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain optical fiber communication with the help of the block diagram. (CO1)

OR

- (b) Describe multimode and single-mode step index fiber with the help of simple ray diagrams. Compare the advantages and disadvantages of these two types of fiber for use as an optical channel. (CO1)
2. (a) A graded index fiber with a parabolic index profile supports the propagation of 742 guided modes. The fiber has a numerical aperture in air of 0.3 and a core diameter of 70 μm . Determine the wavelength of the light propagating in the fiber. Further estimate the maximum diameter of the fiber which gives single-mode operation at the same wavelength. (CO1)

P. T. O.

(2)

TPE-002

OR

- (b) An optical fiber in air has a numerical aperture of 0.4. Compare the acceptance angle for meridional rays with that for skew rays which change direction by 100° at each reflection. (CO1)
3. (a) Discuss absorption losses in optical fibers, comparing and contrasting the intrinsic and extrinsic absorption mechanisms. (CO2)

OR

- (b) Compare stimulated Brillouin and stimulated Raman scattering in optical fibers, and indicate the way in which they may be avoided in optical fiber communications. (CO2)
4. (a) A long single-mode optical fiber has an attenuation of 0.5 dB km^{-1} when operating at a wavelength of $1.3 \text{ }\mu\text{m}$. The fiber core diameter is $6 \text{ }\mu\text{m}$ and the laser source bandwidth is 600 MHz . Compare the threshold optical powers for stimulated Brillouin and Raman scattering within the fiber at the wavelength specified. (CO2)

OR

- (b) Two polarization-maintaining fibers operating at a wavelength of $1.3 \text{ }\mu\text{m}$ have beat lengths of 0.7 mm and 80 m . Determine the fiber birefringence in each case and comment on the results. (CO2)

(3)

5. (a) Describe the phenomenon of modal noise in optical fibers and suggest how it may be avoided. (CO1)

OR

- (b) Determine the cutoff wavelength for a step index fiber to exhibit single-mode operation when the core refractive index and radius are 1.48 and $4.8 \mu\text{m}$, respectively, with the relative index difference being 0.30%.

(CO2)

Roll No.

TPE-004

B. TECH. (ECE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
DIGITAL SYSTEM USING VERILOG

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the various data types available in Verilog with the help of suitable examples. (CO1)

OR

- (b) Draw and explain the VLSI integrated circuit design flow. (CO1)

2. (a) What are the basic components of a module ? Discuss the port connection rules with the help of examples. (CO1)

OR

- (b) Write a Verilog code using gate level modeling for a 4-bit full adder. The adder should take two 4-bit numbers as input and produce a 4-bit sum and a carry output. (CO1)

P. T. O.

(2)

3. (a) Explain system task and compiler directives giving at least two examples of each. (CO2)

OR

- (b) Using Gate-Level modelling write the Verilog code for 2-bit by 2-bit binary multiplier circuit. (CO2)
4. (a) What is a continuous assignment ? Give the syntax of assign statement and explain how delays can be modelled in data-flow modeling. (CO2)

OR

- (b) Evaluate the following, if $A = 1'b1$, $B = 2'd2$, $C = 4'ha$: (CO2)
- (i) $D = \{B, C\}$
 - (ii) $E = D << 1$
 - (iii) $F = \{4 \{D\}\}$
 - (iv) $G = E \neq D$
 - (v) $H = E \& 3'o2$
5. (a) Write the Verilog Code in data-flow modeling for 4-to-1 multiplexer using : (CO2)
- (i) using logic equations
 - (ii) using conditional operators

OR

- (b) Explain the two structured procedure statements in Verilog with one example of each. (CO2)

Roll No.

TPE-007

**B. TECH. (ECE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
SEMICONDUCTOR DEVICES AND MODELING**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Explain C-V characteristics of two terminal MOS structure. (CO1)
OR

(b) Determine the voltage gain, current gain, input impedance and output impedance for common-source amplifier. (CO1)

2. (a) Explain flatband voltage in detail along with expressions. (CO1)
OR

(b) Write short notes on the following : (CO1)
(i) Channel Length Modulation

(ii) Narrow Channel Effects

3. (a) Write short notes on the following :

(i) Body effect in MOS Transistor

(ii) Drain Induced Barrier Lowering (DIBL)

P. T. O.

(2)

OR

- (b) A step graded Ge diode, having $N_D = 500 N_A$, acceptor impurities are 2 impurity atoms for every 10^8 atoms at room temperature. Find the barrier potential. Assume $n_i = 2.5 \times 10^{13} / \text{cm}^3$. Total number of Ge atoms $= 4.4 \times 10^{22} / \text{cm}^3$. (CO2)

4. (a) Explain three terminal MOS structure with proper band diagram. (CO1)

OR

- (b) Calculate the flatband voltage for a p -type body with $N_A = 10^{18} \text{ cm}^{-3}$, a SiO_2 insulator with a thickness $t_{ox} = 2 \text{ nm}$, and a n -type polysilicon gate with $N_D = 10^{20} \text{ cm}^{-3}$. The interface charge $Q_0 = 10^{-8} \text{ C/cm}^2$ and $\phi_{ms} = -1.036 \text{ V}$. (CO1)

5. (a) Explain the effect of gate body voltage of two terminal MOS structure on : (CO2)

- (i) Flatband condition
- (ii) Accumulation
- (iii) Depletion and inversion

OR

- (b) Explain the variation of electron concentration with temperature for n -type semiconductor. (CO2)

Roll No.

TPE-013

B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
SATELLITE COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain the Kepler's law of planetary motion. Briefly discuss the advantages of satellite communication. (CO1)

OR

- (b) A satellite is moving in a circular orbit at a height of 200 km above the surface of earth. Determine its orbital velocity.

Given that $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $m = 5.98 \times 10^{24} \text{ kg}$ and radius of the earth is 6370 km. (CO1)

2. (a) Explain the following with respect to satellite : (CO1)
- (i) Satellite Axis
 - (ii) Ascending Node
 - (iii) Descending Node
 - (iv) Perigee
 - (v) Apogee

P. T. O.

(2)

OR

- (b) The orbital for a satellite orbit has an eccentricity of 0.15 and a semi-major axis of 9000 km. Determine (1) Periodic time, (2) Apogee height and (3) Perigee height.

Assuming earth radius = 6370 km, $\mu = 3.986 \times 10^{14} \text{ Nm}^2/\text{kg}$ (CO1)

3. (a) Explain the different types of orbits. Discuss the Lagrange Points in the Universe. (CO1), (CO2)

OR

- (b) The apogee and perigee of an elliptical satellite orbits are 4000 km and 300 km. Determine the semi-minor axis and semi-minor axis.

(CO1), (CO2)

4. (a) What do you understand by satellite subsystems ? Explain the Altitude and Orbit Control (AOC) Subsystem. (CO2)

OR

- (b) Describe the telemetry, tracking and command (TTC&M) subsystems of a satellite system. (CO2)

5. (a) What is meant by look angle ? Explain them with reference to a geostationary satellite and earth station. (CO2)

OR

- (b) Write a short note on Equipment Reliability. Explain with the help of bath tub curve. (CO2)

Roll No.

TEC-701

**B. TECH. (ECE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

PRINCIPLES OF MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss any *five* goals of management. (CO1)

OR

(b) Mention any *three* definitions of management. (CO1)

2. (a) Discuss interpersonal and informational roles of a manager. (CO1/CO2)

OR

(b) Explain Planning, Organizing, Staffing, Directing and Controlling functions of management. (CO1/CO2)

3. (a) What are various qualities of a manager ? Discuss in detail. (CO1)

OR

(b) Write a short note on internal factors that affect the organization. (CO1)

P. T. O.

(2)

4. (a) Discuss the following levels of management : (CO1/CO2)
top level, middle level and supervisory level.

OR

- (b) Discuss various objectives of planning in detail. (CO1/CO2)
5. (a) Write a short note on strength, weakness, opportunity and threat in an organization. (CO2)

OR

- (b) Discuss all important steps in planning process. (CO2)

Roll No.

TOE-702

**B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
SENSOR NETWORKS (ELECTIVE)**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What is IEEE 802.11 standards ? Elaborate its benefits in network standards. (CO1, CO2)

OR

- (b) Discuss applications of 802.16 protocols. Write its layered structure standards. (CO1, CO2)

2. (a) Discuss Wireless sensor networks and its application. Discuss a suitable example/architecture framework to deploy WSN. (CO2, CO3)

OR

- (b) Discuss major difference in between wireless sensor networks and Ad-hoc Network. (CO2, CO3)

3. (a) Discuss Sink node uses in IOT platform. Discuss its benefits in network topology. (CO1, CO2, CO3)

P. T. O.

(2)

OR

(b) Discuss the application of Microscopic Sensor Nodes in short ?

(CO1, CO2, CO3)

4. (a) Explain Confidentiality and Data integrity in sensor networks.

(CO2, CO3, CO4)

OR

(b) Discuss the Design Goals of WLANs IN brief.

(CO2, CO3, CO4)

5. (a) What is MICA Mote and BT nodes ? Explain its benefits in sensor network structure.

(CO1, CO3)

OR

(b) Discuss in short about Imote and IRIS. Explain its benefits in sensor network structure.

(CO1, CO3)

Roll No.

TEC-731

B. TECH. (ECE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
DISASTER MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Draw the disaster management cycle and explain every stage in it.(CO1)

OR

- (b) Define disaster. Describe natural disasters and man-made disasters with examples of each. (CO1)

2. (a) Explain the application of various technologies in Disaster Management. (CO1/CO2)

OR

- (b) Write short notes on the following : (CO1/CO2)

- (i) Vulnerability
- (ii) Mitigation
- (iii) Early warning system

P. T. O.

(2)

3. (a) What are the factors that affect people's vulnerability ? Explain it.

(CO2)

OR

- (b) Write the roles and responsibilities of community, state and center in disaster mitigation.

(CO2)

4. (a) Enumerate the role of media in disaster management. (CO1/CO3)

OR

- (b) What is disaster awareness ? Write the role of education in risk reduction.

(CO1/CO3)

5. (a) Write a note on Chernobyl Nuclear Disaster and Bhopal Gas Tragedy.

(CO1/CO3)

OR

- (b) Explain how developmental projects are the cause of disaster with proper example.

(CO1/CO3)

Roll No.

TEC-759

B. TECH. (ECE-ESR & IOT) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

INTERNET OF THINGS AND ITS APPLICATIONS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define IoT and characteristics of IoT in details. (CO1)

OR

(b) What are functional block of IoT ? Also explain the communication models for IoT. (CO1)

2. (a) What are the protocols for communication in IoT ? Explain in detail. (CO1)

OR

(b) Explain with the help of IoT model for 'logistics' by fitting the task of logistics to different layers of IoT. How the design can be commercialized and used for society ? (CO1)

P. T. O.

(2)

3. (a) What is the difference between the IoT and M2M ? Explain with the help of suitable examples. (CO2)

OR

- (b) What is Software Define Network ? Explain with the help of suitable diagram. What are the key elements of SDN ? (CO2)
4. (a) A sensor is connected with a microcontroller which is having communication protocols to connect with internet. Draw the possible communication through internet of such device. Can we use wireless sensor network topologies with this device ? Explain your answer with the help of proper diagram and logic. (CO1)

OR

- (b) Explain the outline structure of IoT Architecture. Take example of Agriculture in support of IoT Architecture. (CO2)
5. (a) How does the IoT gateway work ? Explain with the help of suitable diagram. (CO2)

OR

- (b) How can IoT be implemented for Industrial infrastructure ? Explain your example in context of courier service industry. (CO2)